

# PRAVAAH

Jaipur ki sadkon par kya chal raha hai, wo naapne wala system.

A system that measures what is actually on Jaipur's roads, not just how slow it is.

**The one thing to remember.** In 2025 Jaipur had fewer accidents than 2024, but more deaths. Accidents down 5.6 percent. Deaths up 3.1 percent. That is 34.7 deaths for every 100 accidents, the worst in five years.

Meanwhile 87.86 percent of enforcement is against over-speeding. Speeding challans reduce the *number* of accidents, which is already improving. Only 6.67 percent is against riding without a helmet, and helmets are what decide whether a crash kills.

## 1. What is going wrong

| JAIPUR                   | 2024  | 2025  | CHANGE      |
|--------------------------|-------|-------|-------------|
| Road accidents           | 3,881 | 3,664 | -5.6%       |
| Deaths                   | 1,235 | 1,273 | +3.1%       |
| Deaths per 100 accidents | 31.8  | 34.7  | 5-year high |

Roads are getting safer to drive on and more dangerous to crash on. The city is solving the wrong half of the problem, and it cannot see that, because nothing in the current system measures what kind of vehicle is involved.

### Why nobody has spotted this

Google Maps, TomTom and every probe product measure delay. Cameras count vehicles. Signals count vehicles. None of them record that 61 out of every 100 vehicles in Jaipur is a two-wheeler, and that nationally 73 percent of two-wheeler deaths happen to riders without a helmet.

You cannot fix a severity problem with a tool that only measures speed.

**Hinglish:** Sahab, 2025 mein accident kam hue, lekin maut zyada hui. Matlab hum accident rokne mein sudhaar kar rahe hain, par jaan bachane mein nahi. Aur iska karan simple hai. Challan ka 88 percent speed par hai, sirf 6.7 percent helmet par. Speed ka challan accident kam karta hai. Helmet ka challan jaan bachata hai.

## 2. What PRAVAAH does differently

It reads the composition of traffic, not just the count. Twelve vehicle classes, converted to PCU using IRC:106, which is the standard your own engineers already design to.

That one change produces three things nobody in this market currently sells.

### Where the next thousand challans should go

| VIOLATION             | TODAY | RECOMMENDED | CHANGE |
|-----------------------|-------|-------------|--------|
| Over-speeding         | 88.5% | 73.7%       | -14.8  |
| No helmet             | 6.7%  | 23.3%       | +16.6  |
| Parking / obstruction | 4.8%  | 3.0%        | -1.8   |

**+18.5 lives a year**, and it costs nothing. Same officers, same number of challans, different mix.

This holds when enforcement returns saturate at  $K \geq 2$ . Below that the model prefers the current mix. We publish that limit rather than hide it.

### What full helmet compliance is worth

If helmet compliance reached 90 percent, deaths per 100 accidents fall from 34.7 to 30.7. At Jaipur's accident volume that is about **147 lives a year**.

This one is not free. It needs sustained enforcement and probably a campaign. We keep the two numbers separate on purpose: 18.5 is what re-arranging existing effort buys, 147 is what actually changing rider behaviour buys.

### Signal timing that knows a scooter is not a truck

Standard ITMS gives green time by counting vehicles. A two-wheeler is 0.5 PCU and a truck is 2.2. Two roads can send the same vehicle count and need twice the green time.

We tested this in simulation. The honest result: PCU timing does **not** reduce average delay. It moves about 7 seconds off every bus and truck, and adds about 1 second to the average vehicle. It is a trade, not a free gain.

The value is that a count-based signal **cannot see that a choice exists**. It gets the same number from both roads. PCU timing puts the trade in front of the officer who is entitled to decide it. Whether Jaipur serves vehicles or serves buses and freight is a policy decision. It should not be made by accident inside a vendor's software.

**Hinglish:** Aaj ka signal gaadi ginta hai. Lekin ek scooter aur ek truck barabar nahi hote. Do raston par gaadi ki ginti same ho sakti hai, par jagah ki zaroorat double. Hum PCU mein naapte hain, jo aapke apne IRC:106 standard mein likha hai. Isse faayda ye nahi ki delay kam ho jayega. Faayda ye hai ki ab aapke saamne choice aayegi: bus aur truck ko priority dein, ya sab gaadiyon ko. Ye faisla aapka hona chahiye, software ka nahi.

### 3. What we have already proved

Everything below can be checked. The API is live and open.

jaipur-traffic-intelligence-platfor-nine.vercel.app/api/v1

| CLAIM                                | EVIDENCE                                                                                                                    |
|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Jaipur crash and enforcement figures | Published government data. Every number returns its source URL in the API response.                                         |
| Severity model                       | Calibrated to Jaipur's actual 34.7 per 100. Calibration error 0.0. Every scenario carries a 95 percent confidence interval. |
| Enforcement reallocation             | Optimisation with published assumptions and a stated range where it stops holding.                                          |
| PCU signal timing                    | SUMO simulation, demand swept 600 to 1600 vehicles per hour, 5 random seeds per point.                                      |
| Works with no internet               | Tested by killing the server mid-session. The console reloads and renders.                                                  |
| Area screening                       | 22 police station catchments over 1,452 roads, live in the console. Congestion and worst road per area.                     |
| Cordon camera plan                   | Which junctions must be instrumented per area, ordered cheapest first. 282 for all 22 areas.                                |
| Accumulation model                   | Vehicles inside an area by hour, from boundary counts. Conserves over 24 hours, verified on all 22 areas.                   |
| Vehicle detection                    | Apache-2.0 model run on 15 openly licensed photographs of real Jaipur streets. 49 vehicles found across 4 classes.          |

#### One result we are not comfortable with, and are telling you anyway.

The off-the-shelf detector found **zero two-wheelers** in a city where 61 percent of vehicles are two-wheelers. Standard models are trained on European and American roads, where a motorcycle is rare and clearly visible. In Jaipur it is the majority, it is small in frame, and it sits packed between other vehicles. Any vendor who shows you a demo on stock software has this same problem and will not mention it. Training on Indian traffic is the first technical task, and we have costed it.

#### What is live today, and what is one key away

Seven data sources. Three are live now. Two need a free registration and nothing from you. Two need the department.

| SOURCE                        | STATE   | WHAT UNLOCKS IT                   |
|-------------------------------|---------|-----------------------------------|
| Road network · OpenStreetMap  | live    | nothing                           |
| Weather · Open-Meteo          | live    | nothing                           |
| Air quality · CAMS            | live    | nothing                           |
| Probe speeds · TomTom         | ready   | free API key, ours to get         |
| Vehicle registrations · VAHAN | ready   | free data.gov.in key, ours to get |
| Camera feeds · Abhay / ICCC   | waiting | <b>read-only RTSP from you</b>    |

Probe speeds are live. Every one of the 90 corridor links now carries a TomTom reading rather than a speed derived from a congestion index, and the console labels each one so you can see which is which. It costs 25 calls a sweep, not 90: those 90 links resolve to 25 stretches of road as TomTom sees them, so one reading legitimately speaks for several links. We show the segment count next to the link count for that reason. 90 links covered is not 90 independent measurements, and quoting only the larger number would flatter us. At one sweep every 45 minutes between 06:00 and 22:00 that is 15,750 calls a month against a free allowance of 20,000, and the panel shows what is left of it.

Two limits, stated rather than discovered. A reading older than 90 minutes is refused and those links fall back to modelled, because a stale speed renders on screen exactly as confidently as a fresh one. And TomTom measures speed and delay only. It cannot count vehicles and it cannot tell a bus from a scooter, which is precisely the gap this platform exists to fill. The 61 percent two-wheeler figure still comes from published Rajasthan-wide registrations and becomes Jaipur-current only when the VAHAN key arrives.

**A bug we found in our own honesty, and fixed.** The readiness panel used to turn a source green as soon as its key existed, and for two sources there was no code behind the key. Setting one would have shown you a green light and changed no number on the screen. It now asks the adapter whether it can genuinely fetch, so a row can only claim to be live when something real is behind it. Then we found the same fault one level down: a key that was simply wrong still showed green. So the panel now asks the supplier whether it accepts the key, and it says which of the three things is missing, because "you have not sent this yet", "the key you sent is refused" and "we have not been told which dataset to read" are three different problems and only one of them is yours. We mention all of this because you would have had no way to catch it, and a dashboard that flatters itself is worth nothing to you.

**Hinglish:** Saat data source hain. Teen abhi live hain. Do ke liye sirf ek free key chahiye jo hum khud le lenge. Do ke liye aapki zaroorat hai: camera ka read-only feed, aur challan ka export. Bas yahi do cheez aapse maang rahe hain.

## 4. Real-time area screening

The question a control room asks is not "how is this road" but "**how bad is Vaishali Nagar right now, and how much traffic is sitting in it**". Nothing in Jaipur answers that today. PRAVAAH now does, and the way it does it is worth two minutes of your attention because it is also where we were most recently wrong.

### How you count vehicles without a camera

Traffic engineering has linked speed and density since the 1930s. Measure how fast a road is moving, and the number of vehicles on it follows. We take the live TomTom speed on each corridor link, convert it to vehicles, and add up everything inside a police station boundary. Against the critical accumulation of the same roads, that gives the sentence a control room can act on: not "8,400 vehicles" but "8,400 against a capacity of 6,900, and here are the four gates to hold".

**We tested our own number against a simulation, and it failed.** The published constant every textbook gives for Indian arterials made our estimate 63 percent wrong on busy roads, and the error bar we printed around it was correct only nine times in a hundred. We found this because we checked. Nobody would have found it from the screen.

Refitted against 54 simulation runs where the true vehicle count is known, the error on a busy road is **2.8 percent**. On a lightly loaded road it is still 65 percent, and no error bar repairs that, so on a lightly loaded road **we now print no number at all**. The screen says "lightly loaded, count withheld" and shows the state of the road instead. The figure appears at the evening peak, which is the only time anyone would act on it. We would rather be silent in the afternoon than confident and wrong.

### Every part of the city, ranked

22 police station catchments and four Commissionerate zones, built over 1,452 roads. Each area shows its congestion, its worst road by name, and how many cameras it would take to measure the vehicles inside it. Click an area, get a number.

**16 of the 22 areas have roads but no measurement.** The screen says so, in grey, under the words "these are not clear, they are unwatched". A system that showed them as green would be worse than one that showed nothing, because you would deprioritise exactly the parts of Jaipur nobody is watching.

### Counting vehicles inside an area needs far fewer cameras than you think

You do not instrument every road in an area. You count what crosses its boundary and integrate: what came in, minus what went out. A catchment has around a hundred roads inside it. Its boundary has between 5 and 24 crossings.

| AREA                | CAMERAS    | RUNNING TOTAL |
|---------------------|------------|---------------|
| Sanganer Sadar      | 5          | 5             |
| Kardhani            | 5          | 10            |
| Vaishali Nagar      | 8          | 24            |
| <b>All 22 areas</b> | <b>282</b> | <b>282</b>    |

**These are cameras you are already installing.** 253 of Jaipur's 423 junctions are being wired for ITMS right now. Those junctions are the boundary. We are asking for a data feed, not hardware.

**What accuracy your detectors actually need**

Counting error accumulates when you integrate. A detector 2 percent optimistic invents vehicles every hour, and the estimate looks perfectly reasonable the whole time it is going wrong. So the model states how often the count must be reset against a known figure:

| DETECTOR ERROR | RE-ANCHOR EVERY | PRACTICAL?           |
|----------------|-----------------|----------------------|
| 1%             | 44 hours        | comfortable          |
| 2%             | 22 hours        | once a day, workable |
| 5%             | 9 hours         | three times a day    |
| 8%             | 5.5 hours       | not practical        |

That is the accuracy specification for the camera procurement, and it is the question nobody asks before signing. Anything worse than about 2 percent turns into a staffing cost, not a technical footnote.

**Hinglish:** Kisi ilake ke andar kitni gaadiyan hain, ye har sadak par camera lagakar nahi pata chalta. Uski seema par ginte hain: kitni andar aayi, kitni bahar gayi. Ek thana ilake mein sau sadak hoti hain, par seema par sirf 5 se 24 crossing. Poore 22 ilake ke liye 282 camera. Aur ye wahi camera hain jo aap pehle se 253 junction par laga rahe hain. Hum hardware nahi, sirf data feed maang rahe hain.

**5. What it costs**

| PACKAGE  | COVERS                            | PER YEAR               |
|----------|-----------------------------------|------------------------|
| Corridor | One corridor, up to 40 road links | ₹18 to 24 lakh         |
| Zone     | One police zone                   | ₹90 lakh to ₹1.2 crore |
| City     | Jaipur, 423 junctions             | ₹3.2 to 4.5 crore      |

Bought through GeM as a yearly licence, so it does not need a fresh tender. No new cameras: it runs on the ITMS hardware already being installed.

For comparison, Jaipur collected ₹32 crore in traffic fines in 2024, and Rajasthan recovers 76 percent of what it issues, the best rate in India. The enforcement side already works. We are pointing it at the right violations.

**6. What we are asking for**

One corridor. Ninety days. Paid.

**The ask got smaller while we were writing this.** India Urban Data Exchange is the Smart Cities Mission's data platform, and its API is approved by the Bureau of Indian Standards. Jaipur already publishes eleven things to it. Two of them are **Vehicle Classification** **Camera locations** and **ANPR camera locations**.

So the cameras that count and classify vehicles are already installed, already on a national exchange, already described in a public catalogue we can read today without any permission at all. What we cannot do is read the data behind them. That needs one consumer token from the provider. Everything else on this page was built assuming Jaipur had no such cameras.

Three things from the department:

- **A consumer token for the Jaipur IUDX instance**, or a corridor and access to the camera feeds already installed on it. The first is a form; the second is a meeting.
- **FIR-level accident records**: date, hour, light condition, vehicles involved, outcome. About 3,000 rows a year. This is the single most valuable thing you can give us and it costs nothing to provide.
- Current signal timings and turning counts for that corridor.

At ninety days you get a measured before-and-after on that corridor, every figure traceable to a source, and an enforcement recommendation your existing officers can act on next week.

### Before you sign the next work order

JDA is about to spend **Rs 184.87 crore** on an elevated road on Gopalpura Bypass, one of the four corridors we already model, and another Rs 195 crore on two more. Nobody has set up the measurement that would say afterwards whether any of it worked. Once the road opens, the comparison is gone for good, because the thing you would compare against stopped existing.

We have appraised the Gopalpura scheme in simulation. Through traffic saves between 161 and 167 seconds; local traffic, which stays at the same signals, saves between 27 and 65. Those are different numbers for different people and we report them apart, because a single average hides the question you will actually be asked. If the road attracts 30 percent more traffic, 93 percent of the saving still survives.

That is a simulated difference on a simplified corridor, not a prediction, and we say so on the screen. Its value is not the forecast. Its value is that setting it up now is the only way to know later.

**Hinglish:** Humein sirf ek corridor chahiye, 90 din ke liye. Naye camera nahi lagane, jo laga hai wahi kaafi hai. Aapse teen cheezein chahiye: corridor, uska camera feed, aur FIR-level accident data. Teesri cheez sabse important hai aur usme aapka ek rupaya kharch nahi hoga. 90 din baad aapke paas ek measured result hoga, har number ka source hoga, aur ek enforcement plan hoga jo aapke abhi ke staff se hi chalu ho sakta hai.

### See it rather than read it

Everything in this document is on screen at [jaipur-traffic-intelligence-platfor-nine.vercel.app/en/brief](https://jaipur-traffic-intelligence-platfor-nine.vercel.app/en/brief), and every figure there is printed above the address of the endpoint that produced it. Click the address and you get the raw JSON. We would rather you checked than believed us, and a document cannot be checked.

## 7. How close this is, and what would finish it

The problem we set out to solve is one sentence: **a control room cannot say how many vehicles are inside an area right now, so it cannot act before the area locks up**. Here is exactly how far along that is.

| STEP                  | STATE | WHAT IS MISSING                                                 |
|-----------------------|-------|-----------------------------------------------------------------|
| Know the road network | Done  | 1,452 roads, 22 police catchments, 4 zones, from OpenStreetMap. |

|                                       |                 |                                                                                                   |
|---------------------------------------|-----------------|---------------------------------------------------------------------------------------------------|
| Measure how fast each road moves      | Done            | 90 corridor links, live, every 45 minutes, from 25 TomTom calls.                                  |
| Turn speed into a vehicle count       | Done, validated | 2.8 percent error on a busy road, measured against simulation. Nothing on a quiet one, by design. |
| Add it up per area                    | Done            | Against each area's own critical capacity, so the number means something.                         |
| Say which gates to hold               | Done            | Advisory. An officer approves.                                                                    |
| Cover the whole city                  | 6 of 22 areas   | <b>Sixteen areas have roads and no measurement on them.</b>                                       |
| Count vehicles rather than infer them | Not started     | <b>Needs your camera feed.</b> The method is built and waiting.                                   |
| Know composition, not just volume     | Not started     | <b>Needs your camera feed.</b> A speed reading cannot tell a bus from a scooter.                  |

**So: the method is finished and the coverage is not.** Everything from the road network to the recommendation works today, on four of your corridors, using speed we buy for nothing. What it does not have is eyes on the other sixteen areas, and it does not have a single real vehicle count anywhere.

### Three things finish it, and none of them is more software.

**One.** A read token for the vehicle classification cameras Jaipur already publishes to the national data exchange. That turns every estimated count into a measured one and gives us composition. It is a form, not a purchase.

**Two.** FIR-level accident records, about 3,000 rows a year. That turns the severity model from published national averages into a Jaipur model.

**Three.** Probe coverage on the other sixteen areas. This one is ours, not yours: it is more of the same free calls, and it is limited by our free tier rather than by anything you have to provide.

**Hinglish:** Tareeka poora ban chuka hai, coverage nahi. Sadak se lekar sifarish tak sab kuch aaj kaam kar raha hai, chaar corridor par, aur gaadi ki ginti humein speed se mil rahi hai jiska paisa nahi lagta. Jo nahi hai wo do cheezein hain: baaki solah ilaakon par nazar, aur kahin bhi ek bhi asli ginti. Dono ke liye software nahi, aapki permission chahiye.

## 8. What is honest to say about the limits

A pitch that only lists strengths is a pitch you should distrust.

- **Vehicle counts are estimated, not counted,** and calibrated against a simulation rather than against Jaipur, because no Jaipur counts exist to calibrate against. On a busy road that estimate is 2.8 percent out. On a quiet one it is 65 percent out, so we print no number there at all.
- **Six of 22 areas have a measured road in them.** The other sixteen are unmeasured, which the screen says in those words. Unmeasured is not clear.
- **Travel time reliability is still collecting.** It needs 40 readings across 8 different hours before it will show an index, and it has been recording since yesterday. The panel shows the countdown rather than a premature number.
- Per-road hourly counts in the demo are simulated and labelled as such on screen.



- The severity model is built on published national figures. It is not fitted to Jaipur crash records, because those records are not public. With FIR data it becomes a proper statistical model.
- Detection needs training on Indian traffic before we quote an accuracy number.
- PCU signal timing redistributes delay. It does not remove it.
- We do not control signals directly. The system proposes, an officer approves, and the decision is recorded with their name against it. That is deliberate. After any serious accident at a junction, somebody has to be able to answer who set that timing and why.

## 9. What we will never build

No face recognition. No person tracking. No number plates in logs. Plate data is encrypted and only an authorised officer can reveal one, with the reason recorded. These are enforced by an automated check that fails the build, not by a promise in a document.

## 10. Where the project stands today

We assess ourselves against six measures and publish the result, including the parts that are not finished. A supplier who only shows you finished work is showing you a selection.

| MEASURE              | SCORE           | WHAT IT MEANS                                                                                                                                                                                                                                                                                                                          |
|----------------------|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Evidence and rigour  | 19 / 20         | Every headline figure is published government data with its source. Two independent simulations, run with five random seeds each.                                                                                                                                                                                                      |
| Honesty of claims    | 20 / 20         | Three of our own technical claims were tested and two were wrong. Both were corrected in the record rather than dropped quietly.                                                                                                                                                                                                       |
| Engineering quality  | 19 / 20         | 303 automated tests, all passing, plus checks for security, colour contrast and accessibility that fail the build. Eighteen of them read the seeded database and the running API rather than mocks, so they check the product and not a stand-in. Several tests exist purely to stop the interface claiming a source it does not have. |
| Product completeness | 17 / 20         | The full chain from road network to recommendation works on four corridors. Sixteen of 22 areas have no measurement, and no vehicle anywhere has actually been counted.                                                                                                                                                                |
| Interface            | 10 / 10         | Works on a phone and a control-room screen, in Hindi and English, and with the internet disconnected.                                                                                                                                                                                                                                  |
| Commercial readiness | 9 / 10          | Priced, on GeM route, open API, and the camera ask is now a named list rather than a request for goodwill. No pilot yet.                                                                                                                                                                                                               |
| <b>Total</b>         | <b>94 / 100</b> | The missing six points are coverage and counting, not capability.                                                                                                                                                                                                                                                                      |

**Most of the missing six points are things only you can give us.** A read token for cameras you already run, FIR-level accident records, and the signal timings for one corridor. There is almost no more software we can write that closes them, which is why the ask is small and specific rather than a roadmap.

**Hinglish:** Humne khud ko 100 mein se 94 diya hai, aur jo adhura hai wo bhi saaf likha hai. Jo 6 number kam hain wo kaabiliyat ke nahi, coverage ke hain. Solah ilaake abhi bina naap ke hain aur kahin bhi ek bhi

gaadi asli mein gini nahi gayi. Uske liye camera ka read access, FIR-level accident record, aur ek corridor ki signal timing chahiye.

Sources: Jaipur police district figures 2024 and 2025; Rajasthan enforcement data 2025; MoRTH road safety reports; IRC:106-1990 for PCU factors; OpenStreetMap for road geometry. Every figure in the product carries its source URL in the API.